

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv')
```

```
df.head()
```

```
df.info()
```

```
df.describe()
```

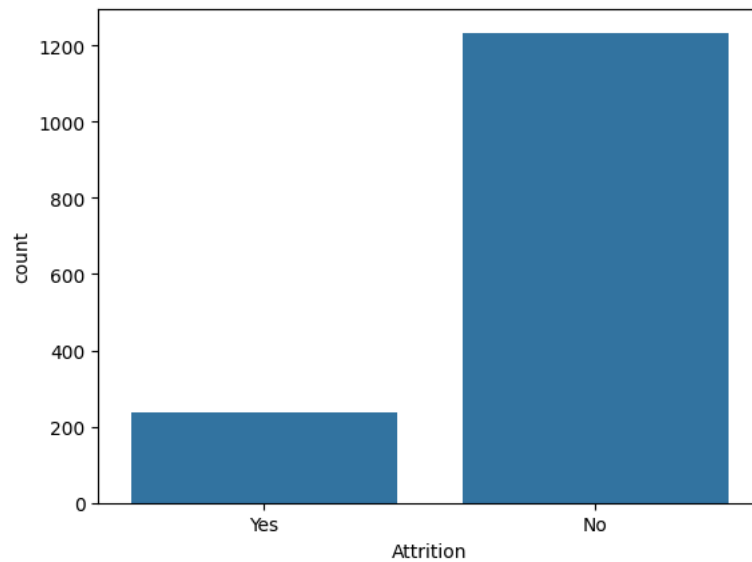
```
df['Attrition'].value_counts()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Age                                  1470 non-null   int64
 1   Attrition                           1470 non-null   object
 2   BusinessTravel                      1470 non-null   object
 3   DailyRate                           1470 non-null   int64
 4   Department                          1470 non-null   object
 5   DistanceFromHome                   1470 non-null   int64
 6   Education                           1470 non-null   int64
 7   EducationField                      1470 non-null   object
 8   EmployeeCount                       1470 non-null   int64
 9   EmployeeNumber                     1470 non-null   int64
10   EnvironmentSatisfaction             1470 non-null   int64
11   Gender                              1470 non-null   object
12   HourlyRate                          1470 non-null   int64
13   JobInvolvement                     1470 non-null   int64
14   JobLevel                           1470 non-null   int64
15   JobRole                             1470 non-null   object
16   JobSatisfaction                     1470 non-null   int64
17   MaritalStatus                      1470 non-null   object
18   MonthlyIncome                      1470 non-null   int64
19   MonthlyRate                         1470 non-null   int64
20   NumCompaniesWorked                 1470 non-null   int64
21   Over18                             1470 non-null   object
22   OverTime                           1470 non-null   object
23   PercentSalaryHike                  1470 non-null   int64
24   PerformanceRating                  1470 non-null   int64
25   RelationshipSatisfaction            1470 non-null   int64
26   StandardHours                      1470 non-null   int64
27   StockOptionLevel                   1470 non-null   int64
28   TotalWorkingYears                  1470 non-null   int64
29   TrainingTimesLastYear              1470 non-null   int64
30   WorkLifeBalance                    1470 non-null   int64
31   YearsAtCompany                     1470 non-null   int64
32   YearsInCurrentRole                 1470 non-null   int64
33   YearsSinceLastPromotion            1470 non-null   int64
34   YearsWithCurrManager               1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

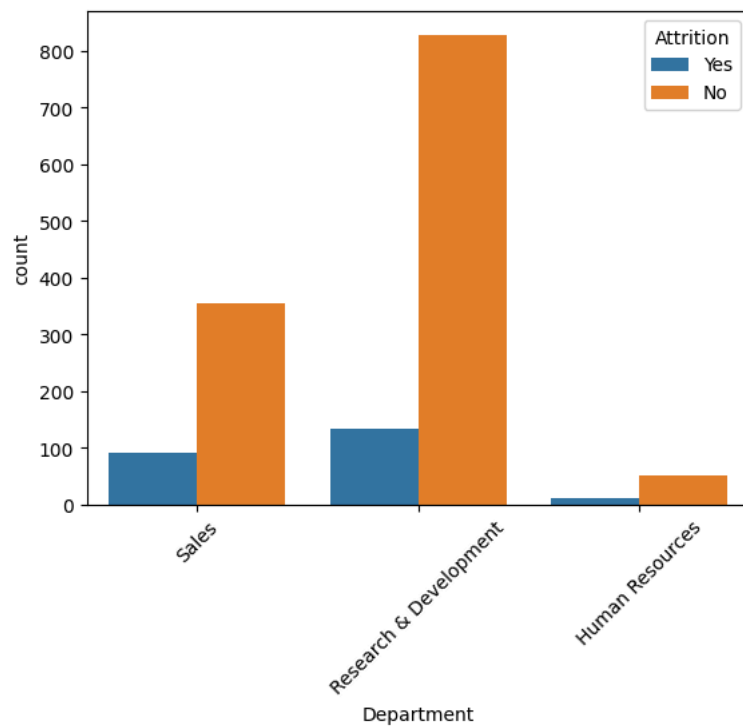
```
count
Attrition
No      1233
Yes      237
```

```
dtype: int64
```

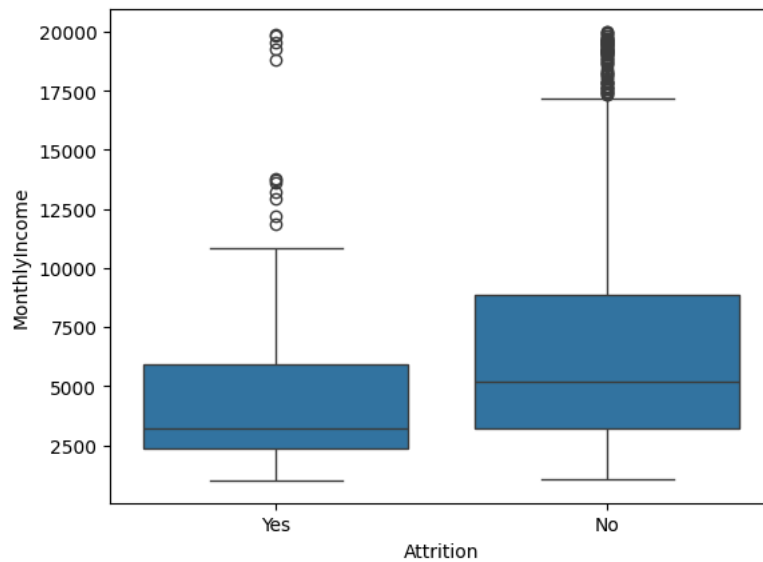
```
sns.countplot(x='Attrition', data=df)
plt.show()
```



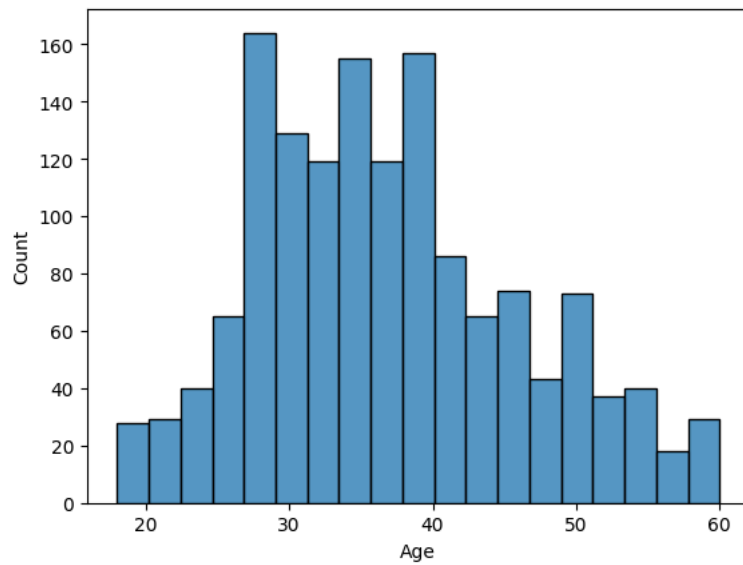
```
sns.countplot(x='Department',  
              hue='Attrition',  
              data=df)  
plt.xticks(rotation=45)  
plt.show()
```



```
sns.boxplot(x='Attrition',  
            y='MonthlyIncome',  
            data=df)  
plt.show()
```



```
sns.histplot(df['Age'])
plt.show()
```



```
df['Attrition'] = df['Attrition'].map({
    'Yes':1,
    'No':0
})
```

```
df = pd.get_dummies(df,
    drop_first=True)
```

```
from sklearn.model_selection import train_test_split

X = df.drop('Attrition',axis=1)

y = df['Attrition']

X_train,X_test,y_train,y_test = train_test_split(
    X,y,
    test_size=0.2,
    random_state=42)
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)

model.fit(X_train,y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
    LogisticRegression
```

```
LogisticRegression(max_iter=1000)
```

```
from sklearn.metrics import accuracy_score
```

```
pred = model.predict(X_test)
```

```
accuracy_score(y_test,pred)
```

```
0.8673469387755102
```

```
from sklearn.metrics import confusion_matrix
```

```
sns.heatmap(
confusion_matrix(y_test,pred),
annot=True)
plt.show()
```

